Scale — Technical-Debt Register

Scope: make every **core** image reference nominally `EntryIdentity`, and let the compiler expose what remains host-typed in core. The edge is explicitly whitelisted.

This register operates at the **host-encoding** level. It is deliberately separate from the *commitments ledger* (`docs/commitments-ledger.typ`), which tracks **cell-semantics** commitments. See also the level model in `types.fs`.

Two axes

Axis	Values	Applies to
Scope	Core / Edge	*where* host types may appear
Level	Cell / Host	*which layer* a type belongs to

A type is legitimate host encoding iff it satisfies both rules:

1. **Projection.** It has a total, injective encoding into tensors and a decoding.
2. **Image-resident state.** It holds no state absent from the image.

A host type that fails either rule has become a *semantic* type and the "Tensor is the only semantic type" invariant is broken.

Nominal types introduced

- **`EntryIdentity`** (Wave 0) — a distinct image reference. `[<Struct>] type EntryIdentity = EntryIdentity of int`, non-negative, unset = `EntryIdentity option`. Projection: rank-0 tensor (an address). Kills the `-1` sentinel.
- **`ValueTag`** (nominal component landed in W3/D14; no host type is needed) — a value's declared sort. **Host encoding only if the sort and the tagging relation are image-resident.** Projection: the name slot of a model entry (a text `EntryIdentity`, or `-1` for anonymous), with declared domains in `DeclarationsRoot`.

Core rows

#	Site	Today	Target	Scope/ Level	Wave	Done
C1	`Lexicon`	`Map<int,Tensor>`	`Map<EntryIdentity,Tensor>`	Core/Host	0	✅ **done** — compiler rejects `int` key
C2	`Image.Root` / `VocabularyRoot`	`-1` sentinel	`EntryIdentity option`	Core/Host	0	✅ **done** — no sentinel; empty round-trips
C3	`Image.addTensor` / `addText`	returns `int`	returns `EntryIdentity`	Core/Host	0	✅ **done** — signature
C4	`Stack`	`Tensor` of doubles-as-indices	`EntryIdentity list`	Core/Host (control state)	0.5	✅ **done** — push/pop round-trip; no doubles-as-indices
C5	`Lace.asIndex` / `asValue`	double decode / first element	`EntryIdentity` rank-0 projection	Core/Host	0.5	✅ **done** — `asIndex` wraps `EntryIdentity`

#	Site	Today	Target	Scope/ Level	Wave	Done
C6	`TextCache`	`Map<string,Map<string,EntryIdentity>>`	`Map<string,Map<string,EntryIdentity>>`	Core/Host	0/1	✅ **done** — values are `EntryIdentity`
C7	`EvaluationStateUserWords`	`Map<string,string>`	`Map<string,string>` resident, keyed by `EntryIdentity`	Core/Host	1	✅ **done** — `UserWordsRoot`; survives persistence
C8	`EvaluationStateBindings`	`Map<string,Tensor>`	`Map<string,Tensor>` resident	Core/Host	1	✅ **done** — `BindingsRoot`; survives persistence
C9	`EvaluationStateModels`	`Map<string,Tensor list>`	`Map<string,Tensor>` resident + sort tag	Core/Host	1 (after `nor`), W3	✅ **done** — `SetsRoot` holds model entries `[nameValue; members…]` (`-1` = anonymous); `DeclarationsRoot` holds declared domains; latest-write-wins, domains grow only. No separate `ValueTag` host type is needed (D14)
C10	`EvaluationState.Program`	`string list`	*re-scoped: control state*	Core/Host	2	✅ **closed (re-scoped)** — see D4
C11	`Tensor.Data`	`double[]`	leaf host storage	Core/Host (leaf)	out	accepted leaf leak
C12	counters (`Location`/ `Position`/ `Baseline`/ `Reductions`)	`int`	out of scope (structural)	—	—	—

`EntryIdentity` is `[<Struct>] type EntryIdentity = EntryIdentity of int`, non-negative, with unset represented as `EntryIdentity option`. `NextIndex` stays `int` (a counter, not a reference). `IndexTransformation` and `Shape` stay `int` (structural, out of scope); `EntryIdentity.value` is the boundary unwrap.

`C4` is reclassified as a **category error** (control state encoded in the data representation), not a host-type leak. `Stack` is allowed host types, the same latitude as the edge. `C10` is likewise **re-scoped as control state** (decision D4): every persistent form of program text (user-word bodies, quotations, bookmark programs) is already image-resident; the live `Program` list is only the interpreter's pending-token work queue and is never persisted.

Wave 2 is closed (re-scoped, no conversion) and **Wave 3 is closed**: the edge track (`Tokenizer`/ `Parser`/`Emitter`/`Bag`) derives its vocabulary from the image (`VocabularyInImage.readVocabulary`) and respects `IsActive`, so edge and core agree on word availability — the live divergence where

`canFly` tokenized and parsed but then failed at evaluation is fixed. The register is now closed; what remains open lives in the commitments ledger as deferred work.

Edge rows (allowed; adapters)

`Tokenizer` string input; `Parser`/`Emitter`/`Bag` string fields and `WordCategory`; `Persistence` `BinaryReader`/`Writer`. Edge rows must **derive from the image copy**, not constitute a second authority (two-sources-of-truth fix).

Named decisions (must precede waves)

1. **Ownership lifetime.** Is linear-ownership state *data* (serialized with the image) or *execution state* (like `Location`, not serialized)? This determines whether C7–C9 need a new serialization section at all. **Answered (D1):** execution state; landed as W9/D30, with no serialization section.
2. **Dictionary encoding.** How an association is represented as image entries — reuse the vocabulary-vector precedent (a vector of `EntryIdentity` pairs), with name resolution via the text index.
3. **Value typing / sorts — landed (W3/D14).** A sort is the image-resident pair `(option Name, Extension)`: the name is a text entry in a model's first slot, the extension is the remaining member identities, and a declared domain lives in `DeclarationsRoot`. `union`/`intersect`/`except` are structural; `merge` is seal-preserving; `complement` reads a declared domain rather than the whole lexicon. The ternary `nor` is unchanged: a binary default-universe form is ambiguous on a stack, so the default landed as a distinct unary word. Substitution-based unification landed in W4/D15 without a new host type or root: terms are image entries (a variable is text, a compound is a `-2` tagged vector, a typed term is a `-3` tagged vector) and the substitution is the existing `BindingsRoot` association vector. `ValueSet.decode` accepts only `-1` as the anonymous marker, so a term cannot be misread as a sort.

Guardrail

If a nominal type ever needs information not projectable into a tensor, that is an alarm — either move the information into the image or revise the invariant. Linear/ownership state is the most likely first violator; its lifetime (data vs. execution state) is therefore also a level decision.

Post-D126 port review

The nine-commit slice (D119–D126) is landed; these are the debts it leaves visible. D127 records the 1288-example / 21-granular verification and the one follow-up code change (`EnableWriter` is Debug-only by default).

- **Writer harness ABI.** Landed for value instructions, stack access, dispatch, and failure (D128): the emitted method takes and returns the real `KernelState<EvaluationState>` and calls the AOT-safe `WriterRuntime`. Units as methods and direct lambda calls are landed (D134); what remains is first-class closures plus the tensor and pattern opcodes.
- **Pattern roles and guards.** Open subjects, `Pin`, `Repeated`, `Shadowing`, and guards remain interpreter-only by D130. They need image-resident bindings and pinned predicate units.
- **Lambda capture.** Capture is conservative (every enclosing name), not a free-variable analysis. A later analysis can shrink captures without changing semantics.
- **Declaration nodes.** `DeclareKind`, `DeclarePredicate`, `Condition`, `Restart`, `Generic`, and `Message` are still named refusals, interpreter-only by D130. They need image-resident registries and a dispatch contract.
- **First adapter.** The first host adapter is C (D130); the conformance harness and version equality land with it. Until then the SageFs suite is the acceptance gate.

- **Sequences and recursive types.** Target (D132), not implemented. Sequences need an ordered projector and words; recursive types need a recursive reference in the portable type document and a fixpoint rule. Multisets and co-products wait for a consumer.
- **Distribution.** Target (D133), not implemented. The `scale_transport_send` ABI seam exists, but node identity, remote send/receive, and partition/failure semantics are undesigned.
- **Capture.** Free-variable capture is implemented (D133); the conservative version is retired.
- **Differential coverage.** The corpus covers the token subset; it does not drive every `lowerExpression` path. Extend it as the lowering grows.
- **Process.** The alpha reload uses SageFs `rebuild=true`, which invokes MSBuild; the owner should confirm this is the intended exception. Typst phrase edits still need `edit` plus `doc_compile`.